

# How Large Language Model Benchmarks Are Built: A Measurement-Oriented Tutorial and Empirical Demonstration with IRT-Viz-Eval

## Abstract

Benchmark scores are widely used to compare large language models, but constructing an LLM benchmark differs from developing an educational or psychological assessment. This tutorial introduces benchmark development to measurement researchers through eight stages, from intended use and evidence specification to validation and maintenance. IRT-Viz-Eval provides a worked example with 84 psychometric figures and 198 tasks. Two vision-language models were administered through Hugging Face Inference Providers and scored against programmatically generated ground truth. Their mean task-normalized scores were nearly identical (.628 and .629), although task profiles, missing outputs, latency, and raw-point aggregation differed substantially. The demonstration shows why benchmark interpretation requires more than a leaderboard total.

**Keywords:** large language models, benchmark design, validity, educational measurement, item response theory

# 1 Why Social Scientists Need a Benchmarking Vocabulary

LLMs are increasingly used in research, education, public administration, and professional work. Social scientists may therefore encounter benchmark scores when selecting a model, evaluating an AI-assisted workflow, or making claims about what a model can do. A model may be described as stronger because it scores higher on a broad knowledge benchmark such as MMLU (Hendrycks et al., 2021), a collaborative task collection such as BIG-bench (Srivastava et al., 2023), or a work-sample evaluation such as SWE-bench (Jimenez et al., 2024). Yet a single number can conceal decisions about content, prompts, sampling, scoring, missing responses, and aggregation.

The social sciences already possess a mature language for asking whether observed performances support intended interpretations. Construct validity, evidence-centered design, reliability, fairness, and consequences are central to educational and psychological assessment (Cronbach and Meehl, 1955; Messick, 1989; American Educational Research Association et al., 2014; Kane, 2013; Mislevy et al., 2003). AI evaluation researchers have begun to apply these ideas to benchmark design. In particular, Evidence-Centered Benchmark Design (ECBD) adapts evidence-centered assessment design to natural language processing and organizes benchmark decisions into capability, content, adaptation, assembly, and evidence modules (Liu et al., 2024). Critical work similarly warns that broad benchmark labels can invite claims that exceed the actual task sample (Raji et al., 2021). This paper builds on that bridge but is written as a practical introduction: what is an LLM benchmark, how is one made, what can go wrong, and what must be reported?

Here, a *benchmark* is a standardized procedure for eliciting observable behavior from one or more AI systems and comparing that behavior with a specified reference, rubric, distribution, or set of desiderata. A benchmark commonly includes data, tasks, prompts, scoring code, aggregation rules, model configuration, and documentation. A leaderboard is

only one possible presentation of its results. A dataset alone is not a complete benchmark because it does not fully specify how a model encounters the data or how outputs become claims.

IRT-Viz-Eval is used as a running case. It asks whether multimodal LLMs can interpret psychometric plots such as item characteristic curves, item and test information functions, test characteristic curves, category response curves, and non-monotonic ideal-point curves. The case is useful because the underlying mathematics provides exact ground truth, while the visual presentation can be manipulated independently. The benchmark itself remains substantively important, but its primary role in this article is to make the benchmark-construction process concrete.

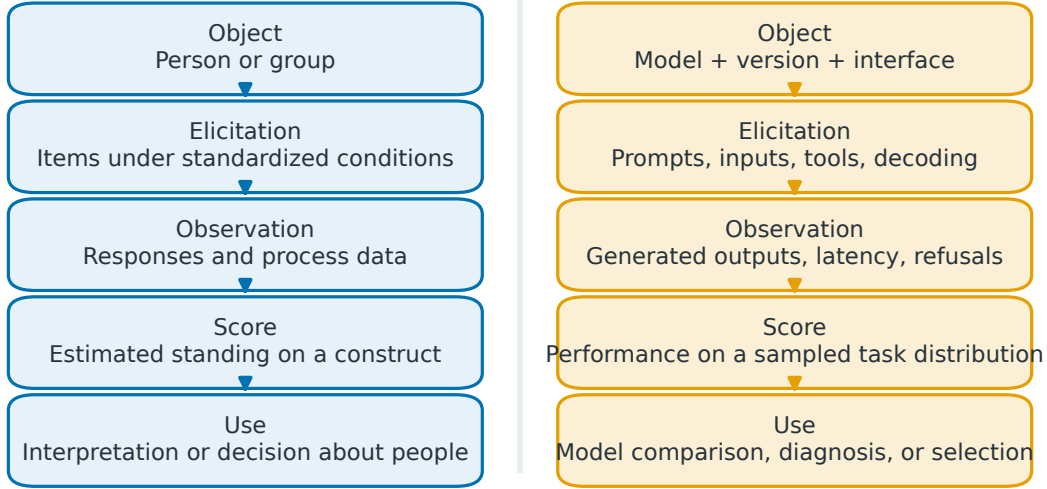
The article is therefore a methodological tutorial with an empirical demonstration, not an attempt to establish a broad model leaderboard. Diagnostic baselines establish that the software behaves as designed; two real model administrations then show how benchmark evidence changes once provider routing, malformed or empty outputs, latency, token limits, and aggregation rules enter the study. The distinction matters: real-model results are necessary for claims about actual systems, but they do not substitute for construct definition, content review, scoring validation, or a defensible interpretation.

## **2 A Benchmark Is Test-Like, but It Is Not a Human Assessment**

The analogy between a benchmark and a test is productive if its limits are explicit. Figure 1 places the two traditions side by side. In both, an object is placed under elicitation conditions, observable responses are scored, and the scores are used to support an inference. The major differences concern what the object is, how stable it is, what “standardization” requires, and what score use is defensible.

## Educational or psychological assessment

## LLM benchmark



Shared principle: observed performance supports an inference only through an explicit validity argument.

**Figure 1:** Parallel anatomy of human assessment and LLM benchmarking. The figure is not claiming that the two enterprises are equivalent. It identifies their shared inferential structure while foregrounding the distinct object and conditions of an LLM benchmark.

## 2.1 The object is a configured system

In most educational and psychological assessments, the object of inference is a person or group. In an LLM benchmark, “the model” is underspecified. Observable behavior may depend on model family, checkpoint or dated version, system message, user prompt, examples supplied in context, image resolution, enabled tools, retrieval corpus, temperature, maximum output length, and provider-side changes. The object is therefore better represented as a configured system: model + version + interface + adaptation condition. A result without this provenance may not be reproducible even when the benchmark items are public. HELM made standardization and release of raw prompts and completions central to transparent model comparison (Liang et al., 2022); model cards and datasheets likewise emphasize documentation of intended use, provenance, and performance conditions (Mitchell et al., 2019; Gebru et al., 2021).

## **2.2 The response process is programmable and potentially stochastic**

Human response processes vary, but a model adds distinctive sources of variability. A generative model can produce different outputs for the same input because of stochastic decoding or provider implementation. It may also return prose when a label was requested, refuse an item, call a tool, or generate malformed structured data. Prompt wording is part of the administration condition, not a neutral wrapper. Even answer-option order can change accuracy on a familiar multiple-choice benchmark (Gupta et al., 2024). Consequently, a benchmark should preserve raw outputs, define parse-failure handling before analysis, and repeat administrations when stochasticity is relevant.

## **2.3 Exposure can invalidate the interpretation**

In conventional testing, advance access to secure items threatens score interpretation. Public LLM benchmarks face an analogous but harder problem: model training corpora are enormous, often undisclosed, and may contain benchmark questions, answers, code, discussions, or task instructions. Contamination can therefore make memorization look like capability (Sainz et al., 2023). Procedures can sometimes detect exposure in black-box models with formal error control, but the general problem cannot be assumed away (Oren et al., 2024). Programmatic generation, newly sampled forms, private holdouts, release dates, and contamination audits are design responses, not complete guarantees.

## **2.4 Scores usually describe a task distribution, not a latent trait**

A human test score is often interpreted as an estimate of standing on a psychological or educational construct, with a measurement model linking items to persons. Many LLM benchmark scores are simpler empirical summaries: average exact-match accuracy, mean rubric points, pass rate, preference rate, or error. They describe performance on sampled

tasks under specified conditions. They do not automatically constitute a latent-variable scale, remain invariant across prompt changes, or support claims about general intelligence. The same benchmark name can also conceal a mixture of qualitatively different tasks. Data composition can affect absolute scores and model rankings, sometimes more than the choice of metric (Kovatchev and Lease, 2024). Scores from benchmarks aimed at similar capabilities may fail to concur (Sun et al., 2023).

## 2.5 The scorer may itself be a model

Closed-form answers can often be scored deterministically. Open-ended generations require exact matching, structured rubrics, expert judgment, crowd judgment, learned metrics, or an LLM-as-a-judge. Model judges can scale evaluation, but they are measurement instruments and require validation. Agreement with human annotation varies markedly across tasks, properties, and judge models (Zheng et al., 2023; Bavaresco et al., 2025). A benchmark should therefore distinguish target model from judge model, preserve judge prompts and versions, and report human agreement or adjudication for a defensible sample.

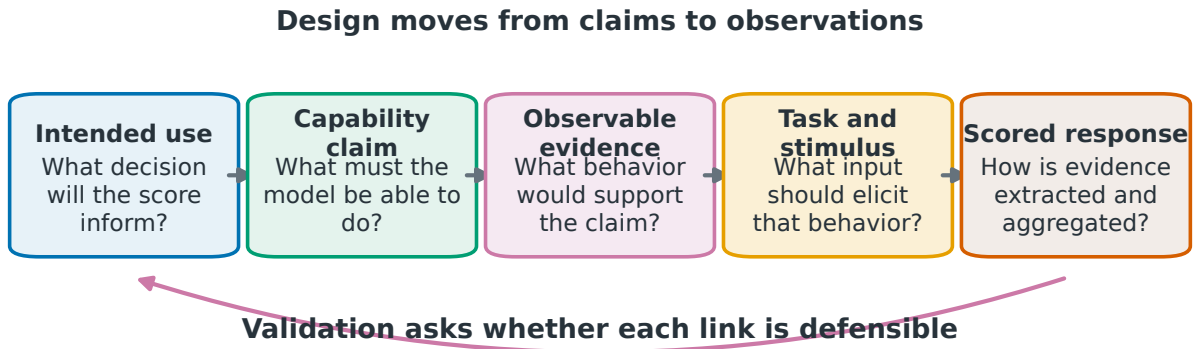
The implication is neither that AI benchmarks are invalid nor that they must copy human testing. Rather, familiar validity principles need to be translated to a different object. Figure 2 summarizes that translation. Design moves forward from an intended use to capability claims, observable evidence, tasks, and scores. Validation works backward and across the chain, asking whether each link is supported.

## 3 How an LLM Benchmark Is Built

Figure 3 presents an eight-stage construction process. It combines the evidentiary logic of ECBD (Liu et al., 2024), the lifecycle perspective of BetterBench (Reuel et al., 2024), and practices from behavioral testing, transparent evaluation, and dataset documentation. The stages are presented sequentially for teaching, although real development is iterative.

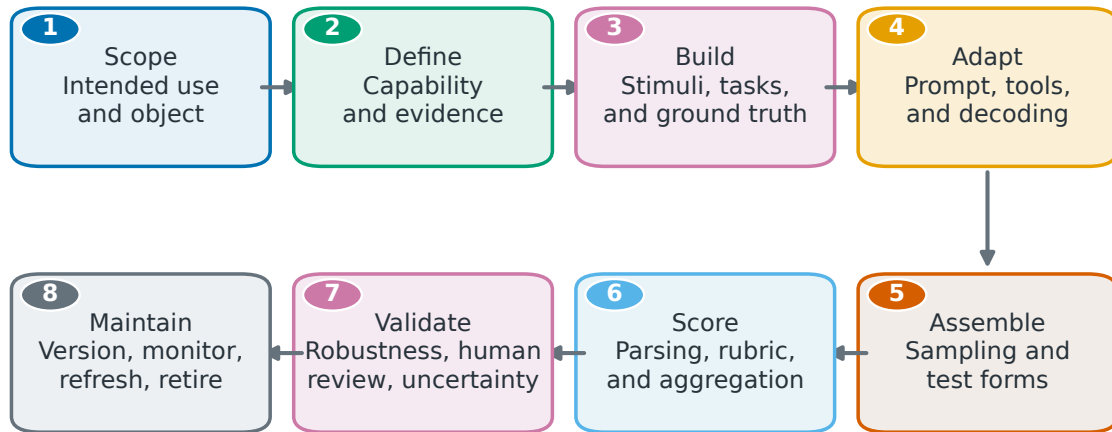
**Table 1:** Practical differences between assessments of people and benchmarks of LLM systems.

| Design issue    | Educational or psychological assessment         | LLM benchmark                                                           |
|-----------------|-------------------------------------------------|-------------------------------------------------------------------------|
| Object          | Person, group, or population                    | Versioned model system under a specified interface                      |
| Administration  | Instructions, timing, setting, accommodations   | Prompt, context examples, media preprocessing, tools, decoding, retries |
| Response        | Selected or constructed human performance       | Generated tokens, tool calls, refusals, latency, malformed outputs      |
| Exposure threat | Item security, coaching, practice               | Pretraining, fine-tuning, evaluation leakage, public prompt reuse       |
| Scoring         | Key, rubric, rater, measurement model           | Parser, executable check, metric, human rater, or model judge           |
| Repeatability   | Affected by persons and occasions               | Also affected by sampling seeds and silent model updates                |
| Typical claim   | Standing on a construct or expected performance | Performance on a sampled task distribution under tested conditions      |
| Lifecycle       | Forms are revised and revalidated               | Public tasks can saturate or contaminate quickly; versions may drift    |



**Figure 2:** The claim-to-evidence chain. A benchmark item is useful only when its response provides evidence relevant to a stated capability, and that capability is relevant to an intended use. Validation examines every link rather than treating the final score as self-interpreting.

## An LLM benchmark is a maintained measurement system, not just a question file



**Figure 3:** An eight-stage benchmark workflow. Scope and construct decisions precede data generation; scoring is designed before claims are made; validation and maintenance are part of the benchmark rather than optional activities after publication.

### 3.1 Stage 1: Specify intended use and object of evaluation

Begin with the decision the benchmark is expected to inform. Candidate uses include scientific diagnosis of model behavior, comparison of models under a controlled research protocol, selection for a specific workflow, monitoring after deployment, or testing a hypothesized failure mode. “Measure AI ability” is too broad. A useful statement names users, systems, conditions, and boundaries. For example: “IRT-Viz-Eval supports researchers and psychometric teams comparing vision-capable models for low-stakes assistance in interpreting clean, static IRT plots. It does not certify unsupervised operational decisions.”

The object of evaluation must be equally explicit. Is it a base model, chat-tuned model, hosted product, retrieval-augmented application, or agent with tools? Can it access the internet? Is the image passed at native resolution? Are prompts zero-shot, few-shot, or optimized separately for each model? A benchmark result applies to this configuration.

## 3.2 Stage 2: Define capabilities and observable evidence

Capability labels should be narrow enough to operationalize. “Reasoning” is not an item specification. “Identify the lower asymptote of a dichotomous item characteristic curve” or “estimate response probability at a marked trait value” is more useful. For each capability, designers should state what a successful output would look like and what competing process could produce the same answer. This is the benchmark equivalent of distinguishing construct-relevant variance from construct-irrelevant variance.

In IRT-Viz-Eval, visual psychometric interpretation is decomposed into model identification, parameter estimation, probability reading, information-peak localization, category-curve reasoning, test-level interpretation, and non-monotonic peak detection. The evidence is not merely a correct label. Numeric estimates and a brief rationale indicate which visible features the model used. The rationale is supportive process evidence, not privileged access to the model’s internal computation.

## 3.3 Stage 3: Construct stimuli, tasks, and ground truth

There are three common content sources. First, experts can author and annotate items. This supports ecological relevance but is expensive and can introduce disagreement. Second, existing datasets can be repurposed. This is efficient but may import poorly documented assumptions and high contamination risk. Third, items can be generated programmatically or by models. This can produce scale, controlled variation, and exact answers, but synthetic regularities may allow shortcuts.

The content specification should cross task types with capabilities, domains, difficulty conditions, and relevant populations or contexts. Behavioral testing is useful here. CheckList distinguishes minimum functionality tests, invariance tests, and directional expectation tests rather than relying only on aggregate held-out accuracy (Ribeiro et al., 2020). IRT-Viz-Eval includes minimum functionality tests such as reading a probability, and invariance tests in which the same curve is rendered under different styles. Figure 4 shows the current content

matrix. Empty cells are informative: they reveal where the benchmark does not provide evidence.

**The content specification crosses visual objects with observable capabilities**

|                                |                     |                |                     |                  |                    |                          |                         |
|--------------------------------|---------------------|----------------|---------------------|------------------|--------------------|--------------------------|-------------------------|
| Psychometric visual object     | Dichotomous ICC     | included       | included            | included         |                    |                          |                         |
|                                | Item information    | included       |                     |                  | included           |                          |                         |
|                                | Polytomous curves   | included       |                     | included         |                    | included                 |                         |
|                                | Test characteristic | included       |                     | included         |                    |                          |                         |
|                                | Test information    | included       |                     |                  | included           |                          |                         |
|                                | Ideal-point curve   | included       |                     | included         |                    |                          | included                |
|                                |                     | Identify model | Estimate parameters | Read probability | Locate information | Reason across categories | Detect non-monotonicity |
| Capability sampled by the task |                     |                |                     |                  |                    |                          |                         |

**Figure 4:** IRT-Viz-Eval content specification. Rows define psychometric visual objects; columns define observable capabilities. Green cells identify included task combinations. The matrix makes both coverage and omissions visible before a total score is calculated.

Ground truth should be generated or annotated independently of the target model whenever possible. Executable tasks can be checked by running code or tests, as in SWE-bench (Jimenez et al., 2024). Quantitative figures can be linked to their underlying data table; ChartQA, for example, explicitly studies visual and logical reasoning over chart images and data (Masry et al., 2022). IRT-Viz-Eval stores generating parameters and coordinates alongside each image, allowing exact answers to be derived rather than judged impressionistically.

### 3.4 Stage 4: Standardize adaptation and elicitation

An LLM rarely consumes a task without adaptation. The benchmark must specify the system prompt, user prompt, examples, output schema, media transformations, tool access, and

decoding parameters. Adaptation can be standardized across models for comparability or tailored to each model for best-achievable performance; these answer different questions. A fair study may report both.

Prompts should be piloted for comprehension and unintended cues. Structured output simplifies scoring but creates format-following demands. If JSON is required, researchers should report malformed-output rates rather than silently discarding them. When prompts are varied, results should be treated as a distribution over plausible administrations. Changes to answer order, formatting, or labels can function as invariance checks rather than nuisances (Gupta et al., 2024).

### **3.5 Stage 5: Assemble forms and choose a sampling distribution**

Benchmarks often conflate the item pool with the evaluated form. The assembly step determines which items a model actually receives and therefore what the score weights. Sampling should reflect the intended claim. Equal numbers per category estimate a balanced construct profile; prevalence-weighted sampling estimates performance under a target use distribution. Both are legitimate, but they are not interchangeable.

Designers should specify exclusions, duplicate handling, train-development-test separation, random seeds, and whether all models receive identical forms. A benchmark should be large enough to make relevant comparisons stable, but item count alone does not guarantee coverage. BetterBench found that uncertainty reporting and easy replication were uncommon even among established benchmarks (Reuel et al., 2024). Bootstrap intervals over items, repeated runs, and hierarchical models can separate item, model, and administration variation.

### **3.6 Stage 6: Score, aggregate, and preserve provenance**

Scoring begins with the response representation. Exact matching is transparent but brittle. Numeric tolerance bands recognize approximate visual reading. Executable validation is strong when a task has a deterministic success condition. Human rubrics support nuanced

outputs but require rater training and agreement analysis. LLM judges reduce cost but need task-specific human validation (Bavaresco et al., 2025).

Aggregation requires substantive justification. A mean can allow strength in one domain to compensate for failure in another. A worst-group or minimum-capability score encodes a different use. Accuracy and efficiency may conflict. HELM’s multi-metric design illustrates why performance, robustness, fairness, calibration, toxicity, and efficiency should not automatically collapse into one rank (Liang et al., 2022). At minimum, report total and disaggregated results, denominators, uncertainty, parse failures, refusals, and cost or latency where relevant.

### **3.7 Stage 7: Validate the benchmark and its interpretation**

Validation is an accumulating argument, not a one-time coefficient. Useful evidence includes expert content review, item-level error analysis, comparison with human judgments, expected ordering of diagnostic baselines, robustness to nuisance transformations, convergence or divergence with related benchmarks, and prediction of performance in the intended workflow. Disagreement with another benchmark is not automatically a flaw, but it requires an account of why the constructs or task distributions differ (Sun et al., 2023).

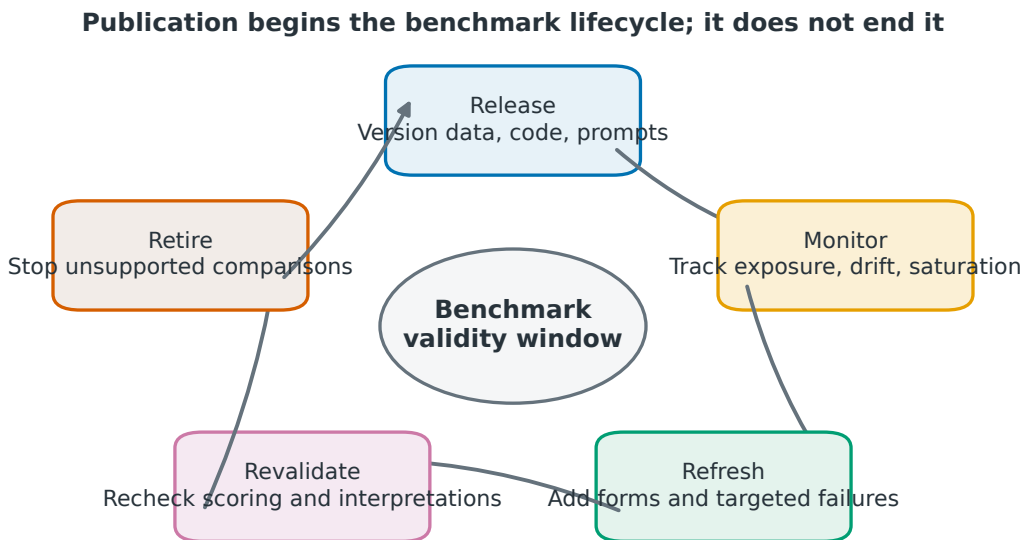
Benchmark quality should itself be tested. Does an oracle obtain the ceiling? Does a deliberately corrupted response receive partial credit? Does a blind response perform near a meaningful floor? Do small visual changes that preserve the answer leave scores stable? Do changes intended to increase difficulty reduce performance? These checks resemble unit tests, simulation studies, and known-groups validity evidence.

### **3.8 Stage 8: Maintain, refresh, and retire**

A public benchmark changes after release even if its files do not. Items can enter model training corpora, developers can optimize against the leaderboard, tasks can saturate, provider versions can drift, and intended uses can change. Dynabench treats data collection and

evaluation as a dynamic human-and-model-in-the-loop process (Kiela et al., 2021). A less resource-intensive project can still version releases, retain private or newly generated forms, publish change logs, monitor ceiling effects, and state retirement criteria.

Figure 5 makes this lifecycle explicit. A benchmark has a validity window: a period during which its evidence supports stated comparisons under documented conditions. Release is therefore the start of monitoring, not the endpoint of design.



**Figure 5:** Benchmark maintenance cycle. Monitoring addresses exposure, model drift, and saturation; refreshes require renewed validation; retirement prevents unsupported historical comparisons from being treated as current evidence.

## 4 Worked Example: Constructing IRT-Viz-Eval

### 4.1 Intended use and capability model

IRT-Viz-Eval is intended to support research on whether vision-capable language models can interpret visual artifacts used in educational measurement. It is not a certification exam for psychometricians and does not support autonomous high-stakes use. Its present domain is

clean, static plots with known generating parameters. The capability claim is correspondingly bounded: under a documented prompt and image input condition, a model can recover psychometrically relevant properties from an IRT figure and explain the visible evidence for its answer.

The substantive domain includes dichotomous 1PL, 2PL, 3PL, and 4PL item characteristic curves (ICCs); item information functions (IIFs); partial credit, generalized partial credit, and graded response category curves; test characteristic curves (TCCs); test information functions (TIFs); and a transparent non-monotonic ideal-point family. These objects have visible signatures connected to established IRT parameters (Rasch, 1960; Birnbaum, 1968; Lord, 1980; Samejima, 1969; Andrich, 1978; Muraki, 1992; Roberts et al., 2000).

## 4.2 Programmatic content and factorial rendering

All stimuli are generated on a shared ability grid from  $\theta = -4$  to 4. Dichotomous response probabilities use

$$P_i(\theta) = c_i + \frac{d_i - c_i}{1 + \exp[-1.7a_i(\theta - b_i)]}, \quad (1)$$

where  $a_i$  is discrimination,  $b_i$  is difficulty,  $c_i$  is the lower asymptote, and  $d_i$  is the upper asymptote. Item information is

$$I_i(\theta) = \frac{[P'_i(\theta)]^2}{P_i(\theta)[1 - P_i(\theta)]}. \quad (2)$$

TCCs and TIFs sum item response and information functions. Polytomous curves are produced from PCM, GPCM, and GRM probability functions. The non-monotonic stress test uses

$$P_i(\theta) = l_i + (u_i - l_i) \exp \left[ -\frac{1}{2} \left( \frac{\theta - \delta_i}{\sigma_i} \right)^2 \right]. \quad (3)$$

This final family is an interpretable ideal-point curve rather than a full GGUM implemen-

tation; its purpose is to test detection and localization of non-monotonicity.

Programmatic generation separates mathematical content from presentation. As Figure 6 shows, the local validation contained 14 base mathematical stimuli. Each was rendered under six style profiles, creating 84 images and 198 linked tasks. This factorial structure makes it possible to compare responses to the same mathematical object under changed visual conditions.

**Programmatic generation separates mathematical content from visual presentation**



A full local pipeline run then created 594 responses and judgments across three diagnostic baselines.

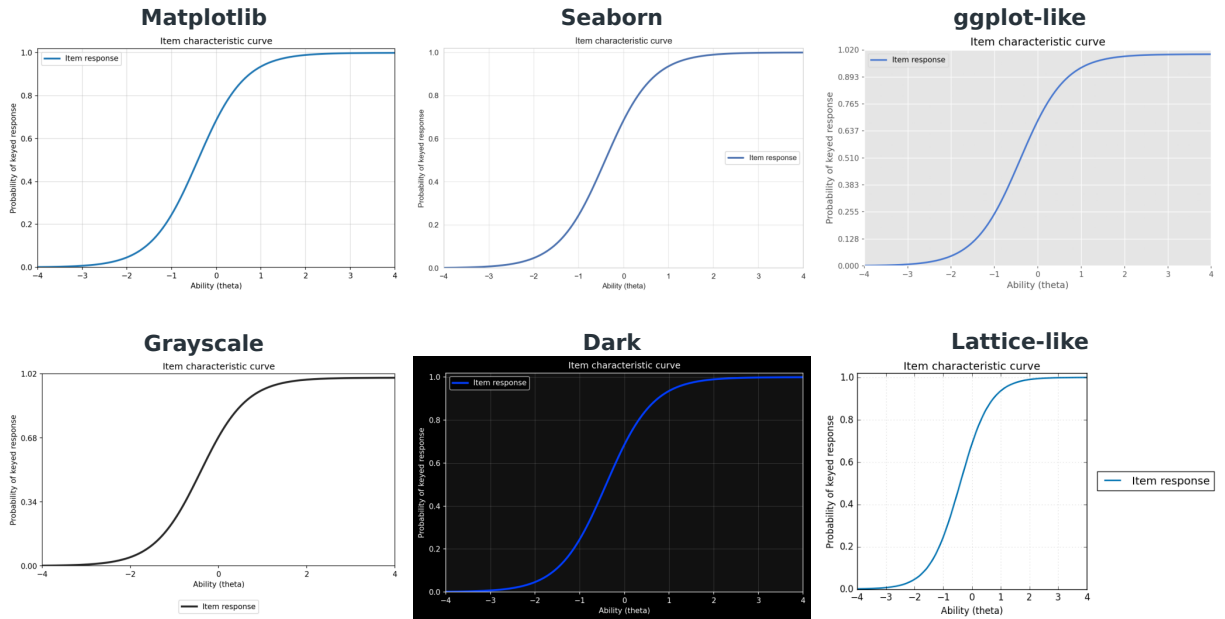
**Figure 6:** Expansion of the local IRT-Viz-Eval validation set. The multiplication by rendering profile changes visual presentation while preserving the underlying parameters and expected answers. Different images generate different numbers of tasks, producing 198 total task records.

### 4.3 Invariance as a designed source of evidence

Six profiles vary background, palette, gridline strength, tick density, legend placement, and line width: Matplotlib light, Seaborn white-grid, ggplot-like gray, grayscale without a grid, dark high-contrast, and lattice-like. Figure 7 displays the same 2PL ICC in all six conditions. Because the mathematical curve is held fixed, a drop associated with style is interpretable as a visual robustness failure rather than a harder IRT item. This is a benchmark analogue of an invariance test (Ribeiro et al., 2020).

The current styles are controlled perturbations, not a representative sample of all real psychometric software. Future forms should add confidence bands, empirical noise, dense

**The mathematics is fixed while the rendering changes**



**Figure 7:** Six renderings of one underlying 2PL item characteristic curve. The curve coordinates and ground truth are identical. Variation in model performance across panels therefore diagnoses sensitivity to presentation rather than psychometric content.

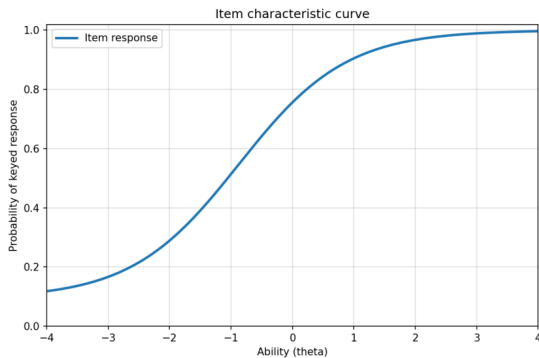
multi-item overlays, mislabeled or truncated axes, grayscale printing artifacts, and plots exported from operational systems. These additions would extend the claim from clean synthetic figures toward the figures encountered in practice.

#### 4.4 One benchmark instance as an evidence record

Figure 8 shows why an item image alone is not the benchmark. A complete instance links the image to a standardized prompt, requested response schema, expected fields, field-specific thresholds, provenance, and raw output. The example asks for four dichotomous item parameters. The target output is compact JSON plus a rationale. The benchmark stores both parsed values and the original output so scoring decisions can be audited.

##### An individual benchmark instance is a complete evidence record

#### 1. Stimulus shown to the model



#### 2. Standardized elicitation

Estimate a, b, c, and d from the item characteristic curve.  
Return JSON with the requested fields and a short rationale.

#### 3. Structured observation and scoring

```
{"a": 1.42, "b": -0.36,  
 "c": 0.18, "d": 1.00}
```

Compare each field with stored ground truth.  
Retain parsing failures, rationale score,  
model version, prompt, and latency.

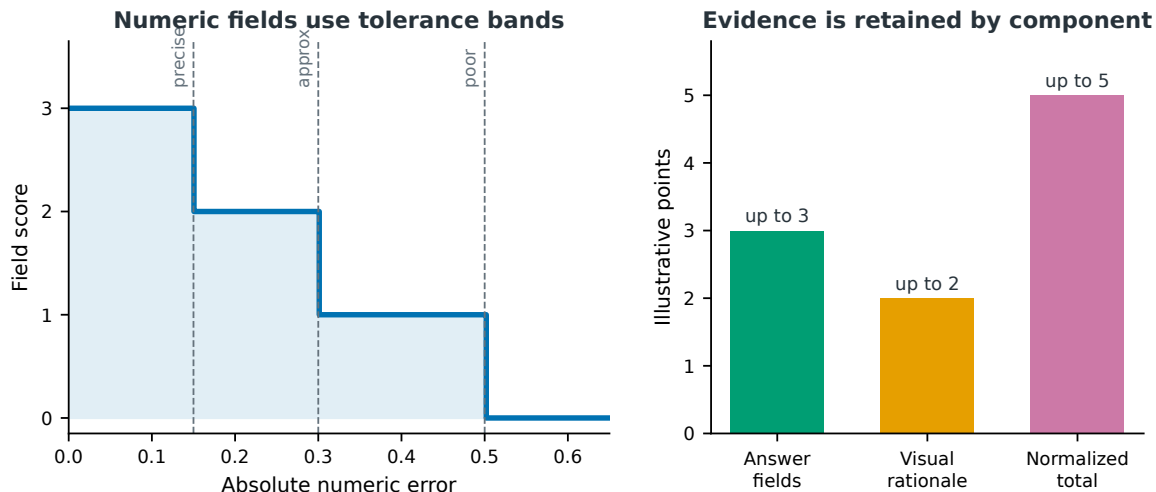
**Figure 8:** Anatomy of an IRT-Viz-Eval instance. The stimulus, elicitation, structured observation, ground truth, and provenance together form the unit of evidence. Keeping these links prevents an image file or a score from becoming detached from its administration conditions.

The repository writes four linked JSONL files. A stimulus record contains the base stimulus identifier, curve family, IRT model, parameters, coordinate path, image path, and visual metadata. A task record contains the prompt, expected fields, and thresholds. A response record contains model identity, decoding settings, latency, raw output, and parsed output. A judgment record contains field-level scores, rationale score, total, normalized score, and judge provenance. JSON schemas permit automated validation and make missing provenance visible.

## 4.5 Scoring design

Categorical and integer fields receive exact-match credit after conservative normalization. Numeric fields use task-specific absolute-error bands. In the common parameter example, errors within 0.15, 0.30, and 0.50 receive 3, 2, and 1 points; larger errors receive 0. Probability-reading tasks use tighter bands of 0.05, 0.10, and 0.20. A deterministic rationale rule adds zero to two points based on the presence of prespecified psychometric terms and minimum explanatory content. This component is a reproducible proxy for visible rationale evidence, not a semantic judgment of whether every explanation is correct. Scores are normalized by the maximum available points for that task because task families request different numbers of fields.

Figure 9 emphasizes two choices. First, visual reading is approximate, so a graded numeric rubric is more informative than exact string equality. Second, the normalized total is not the only result. Field errors, rationale indicators, format compliance, and task type remain available for diagnosis. A future validation study should ask experts to score a stratified response sample and estimate agreement with the deterministic rubric before replacing or supplementing it with human or LLM judgment.



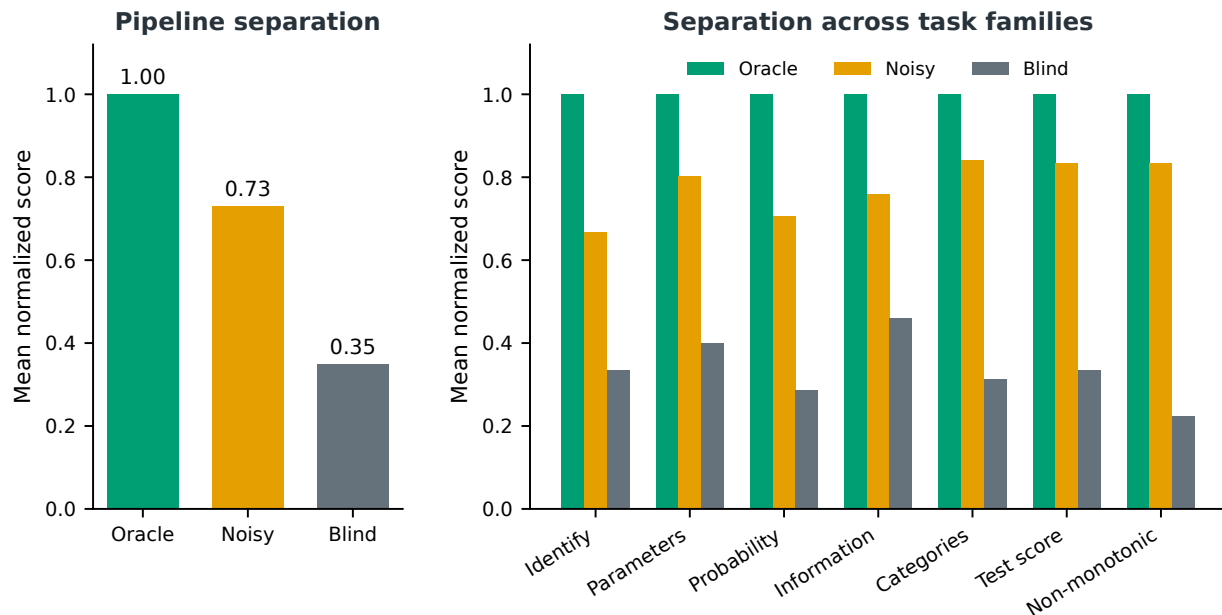
**Figure 9:** Illustration of IRT-Viz-Eval scoring. The left panel shows tolerance-band scoring for a numeric field. The right panel shows why component scores are retained before normalization: answer accuracy and rationale evidence can be inspected separately. Point totals are illustrative because the number of answer fields differs by task.

## 5 From Benchmark Prototype to Actual Model Administration

### 5.1 What the local baselines establish

The local validation used seed 20260711 and one sampled base instance per model family. It produced 84 rendered stimuli and 198 tasks. Three deterministic baselines generated 594 responses: an oracle copied expected answers, a noisy baseline perturbed correct answers, and a blind baseline returned generic guesses. These are software diagnostics, not language models and not estimates of real-world performance.

The expected ordering appeared overall and within every task family. The oracle mean normalized score was 1.00, the noisy baseline was 0.73, and the blind baseline was 0.35. Figure 10 shows the full profile. The result establishes that generation, parsing, field scoring, normalization, and aggregation distinguish exact, approximate, and weak responses. It does not establish content validity, human-model correspondence, or deployed-model competence.



**Figure 10:** Diagnostic pipeline validation. The left panel shows overall separation of the oracle, noisy, and blind baselines; the right panel shows separation by task family. Because these are constructed local baselines, the graph validates benchmark mechanics only.

## 5.2 Choosing an API route

Real-model responses are necessary when a claim concerns an actual system rather than benchmark mechanics. The route used to obtain those responses is part of the administration condition. Table 2 summarizes common options. Direct vendor APIs provide model-native controls; routing services expose models from several developers through one interface; dedicated endpoints provide a controlled hosted deployment; and local inference offers the greatest implementation control when hardware and open weights are available. OpenAI’s Responses API and Google’s Gemini API both accept image inputs, while Hugging Face Inference Providers routes requests to supported third-party models and compute providers through a common client (OpenAI, 2026; Google, 2026; Hugging Face, 2026).

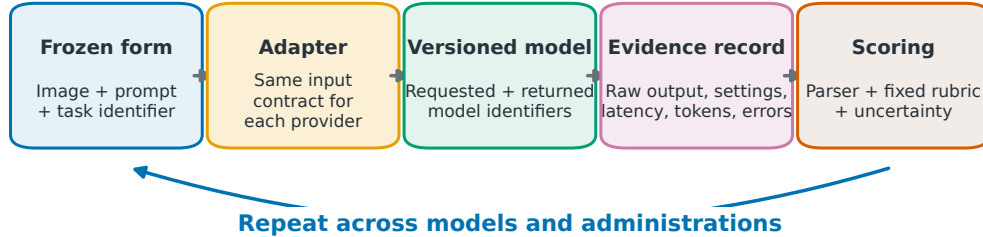
Figure 11 shows the common administration path. Each task record supplies an image, prompt, task identifier, output expectation, and scoring rule. A provider adapter sends the same model-visible content to each target system. The resulting evidence record stores the requested and returned model identifiers, access time, output limit, repetition number,

**Table 2:** API and deployment options for administering a multimodal benchmark.

| Route               | Examples                               | Primary advantage                                 | Measurement concern                                                     |
|---------------------|----------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------|
| Direct vendor API   | OpenAI Responses; Google Gemini        | Native model controls and provider metadata       | Separate credentials, billing, schemas, and version policies            |
| Model router or hub | Hugging Face Inference Providers       | One task adapter can reach several model families | Routing and underlying compute provider must be recorded                |
| Dedicated endpoint  | Managed cloud or Hugging Face endpoint | Stable deployment and configurable resources      | Setup cost and endpoint configuration become part of the system         |
| Local inference     | Transformers, vLLM, or similar runtime | Control over weights, decoding, and logs          | Hardware, quantization, runtime, and preprocessing affect comparability |

latency, token usage, raw output, and any API or parsing error. The scorer operates on these stored records rather than on a live chat transcript.

#### Administering a benchmark to actual models is a controlled study, not a manual chat



Use diagnostic baselines to test mechanics; use real model calls only when making claims about model performance.

**Figure 11:** Reproducible administration of IRT-Viz-Eval to actual models. Consumer product labels such as “ChatGPT” are insufficient provenance; a study should record the API model identifier and returned version whenever available. Diagnostic baselines and real model calls answer different questions and should remain separate.

The repository implements both an OpenAI multimodal Responses API adapter and the Hugging Face adapter used in this study. The latter used Hugging Face authentication and

routing but explicitly selected the underlying inference provider in each model identifier. This is an important distinction: Hugging Face supplied the common interface, model discovery, authentication, and request routing; Novita served GLM-4.5V and DeepInfra served Gemma 3 4B. A consumer subscription to ChatGPT or Gemini is not an API administration record and does not by itself supply API credits. Manual consumer-chat administration was avoided because routing, hidden instructions, media preprocessing, and versioning are difficult to audit.

### 5.3 Empirical administration protocol

The benchmark was administered on July 11–12, 2026. The two target systems were requested as `zai-org/GLM-4.5V:novita` and `google/gemma-3-4b-it:deepinfra`; the router returned `zai-org/glm-4.5v` and `google/gemma-3-4b-it`, respectively. Both received all 198 zero-shot tasks, the same PNG images and prompts, no tools, and a maximum of 1,200 completion tokens. Each prompt requested task-specific JSON plus a brief visual rationale. Unspecified decoding parameters used provider defaults. Calls used a 90-second timeout, one retry for transient network failures, resumable JSONL output, and one administration per task. Table 3 records the core conditions.

No LLM was used as a judge. Categorical fields were scored by normalized exact match, numeric fields by prespecified tolerance bands, and rationale evidence by the fixed rule described earlier. The primary model score was the unweighted mean of the 198 task-normalized scores. To represent uncertainty over sampled mathematical content, 95% intervals were calculated from 10,000 cluster-bootstrap draws over the 14 base stimuli. The paired interval for the model difference resampled the same base-stimulus clusters. These intervals do not represent within-model sampling variability because only one response per task was collected.

**Table 3:** Frozen conditions for the two Hugging Face-routed administrations.

| Condition                 | GLM-4.5V                                                                                                 | Gemma 3 4B                                  |
|---------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------|
| Requested identifier      | <code>zai-org/GLM-4.5V:novita</code>                                                                     | <code>google/gemma-3-4b-it:deepinfra</code> |
| Returned identifier       | <code>zai-org/glm-4.5v</code>                                                                            | <code>google/gemma-3-4b-it</code>           |
| Router / compute provider | Hugging Face / Novita                                                                                    | Hugging Face / DeepInfra                    |
| Tasks and repetitions     | 198 tasks; one response per task                                                                         | 198 tasks; one response per task            |
| Input and elicitation     | Native benchmark PNG; zero-shot task prompt; JSON requested; no tools                                    | Same                                        |
| Output and network policy | 1,200 tokens; 90-second timeout; one transient retry                                                     | Same                                        |
| Scoring                   | Deterministic field rubric plus rule-based rationale component; empty or unparseable fields receive zero | Same                                        |

## 5.4 Two-model results

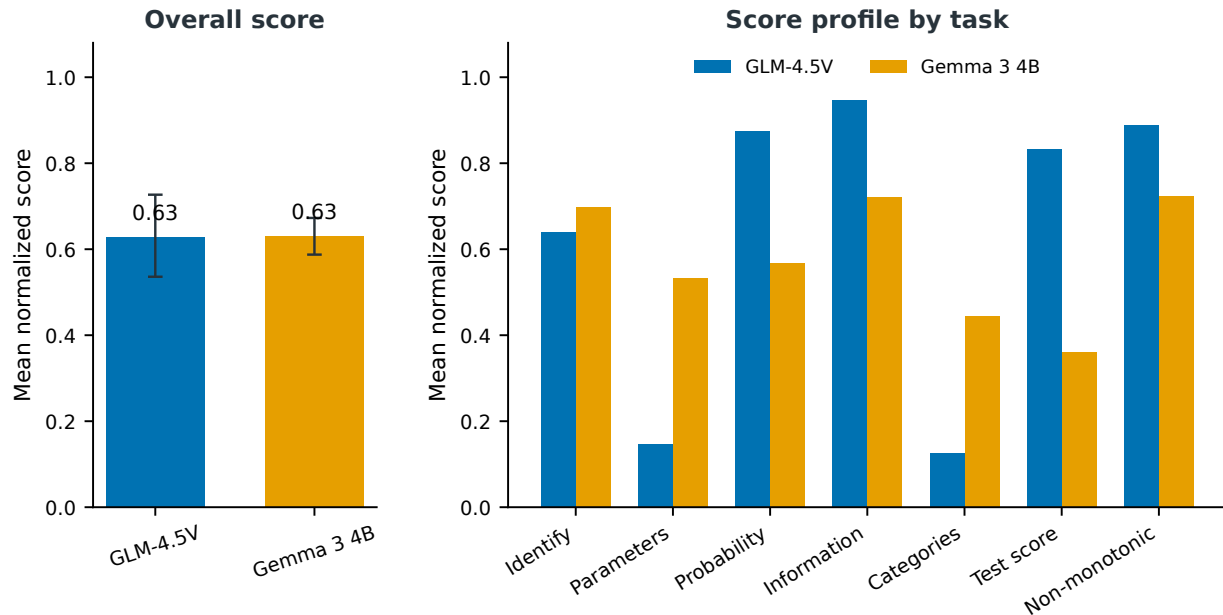
Both models completed 198 API calls without a recorded API error after deduplication. Their primary scores were almost identical: .628 for GLM-4.5V (95% cluster-bootstrap CI [.536, .727]) and .629 for Gemma 3 4B (95% CI [.587, .673]). The paired difference, GLM minus Gemma, was -.001 (95% CI [-.078, .076]). The design therefore does not support an overall ordering of these systems. Table 4 also shows a consequential aggregation choice. Gemma received more raw rubric points, 742 of 1,236, than GLM, 668 of 1,236, although their mean task-normalized scores were nearly equal. Raw-point aggregation gives more weight to tasks with more requested fields; task normalization gives every task equal weight.

**Table 4:** Overall scores and administration diagnostics for the complete 198-task runs.

| Model      | Mean normalized | 95% CI       | Raw points | Empty output | Median latency |
|------------|-----------------|--------------|------------|--------------|----------------|
| GLM-4.5V   | .628            | [.536, .727] | 668/1,236  | 16.7%        | 13.79 s        |
| Gemma 3 4B | .629            | [.587, .673] | 742/1,236  | 0.0%         | 2.14 s         |

Operationally, the two runs were not equivalent. GLM produced visible text for 165 tasks and parseable JSON for 152; Gemma produced nonempty, parseable JSON for all 198.

GLM’s median latency was 13.79 seconds (interquartile range 8.42–21.93), compared with 2.14 seconds (1.75–2.53) for Gemma. Provider-reported total token counts were 342,107 and 85,531, respectively. These values describe the routed systems under this protocol, not immutable properties of the model weights: providers can differ in image tokenization, reasoning-token accounting, serving hardware, and defaults.



**Figure 12:** Complete two-model results. Overall bars show mean task-normalized scores with 95% cluster-bootstrap intervals over base mathematical stimuli. Task-level bars reveal different capability profiles despite nearly identical overall means.

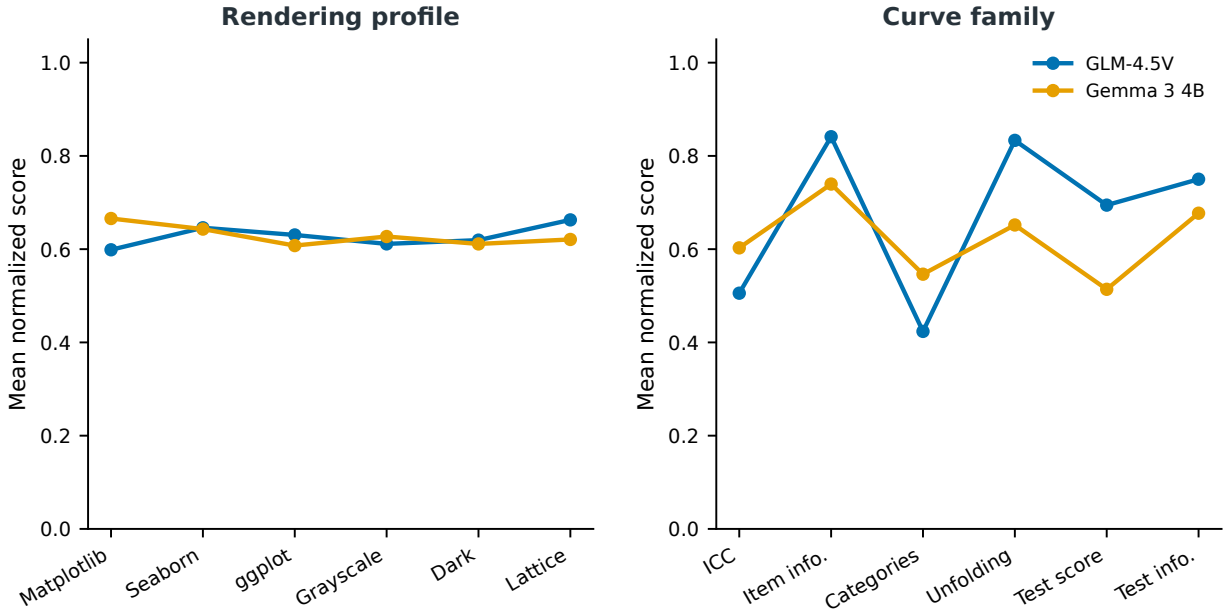
The overall tie concealed substantial profile differences (Table 5; Figure 12). GLM was stronger on information-peak localization (.946 vs. .721), probability reading (.873 vs. .567), non-monotonic peaks (.889 vs. .722), and test expected scores (.833 vs. .361). Gemma was stronger on parameter estimation (.533 vs. .147), category-curve reasoning (.444 vs. .125), and model identification (.698 vs. .639). These are descriptive comparisons from one administration; they identify hypotheses for repeated evaluation rather than stable model traits.

Figure 13 disaggregates by rendering profile and curve family. Style scores varied within a comparatively narrow range for each model: .599–.663 for GLM and .608–.666 for Gemma.

**Table 5:** Mean normalized score by task type.

| Task type                   | $n$ | GLM-4.5V | Gemma 3 4B |
|-----------------------------|-----|----------|------------|
| Model identification        | 84  | .639     | .698       |
| Parameter estimation        | 24  | .147     | .533       |
| Probability at $\theta = 0$ | 30  | .873     | .567       |
| Information peak            | 30  | .946     | .721       |
| Category-curve reasoning    | 18  | .125     | .444       |
| Test expected score         | 6   | .833     | .361       |
| Non-monotonic peak          | 6   | .889     | .722       |

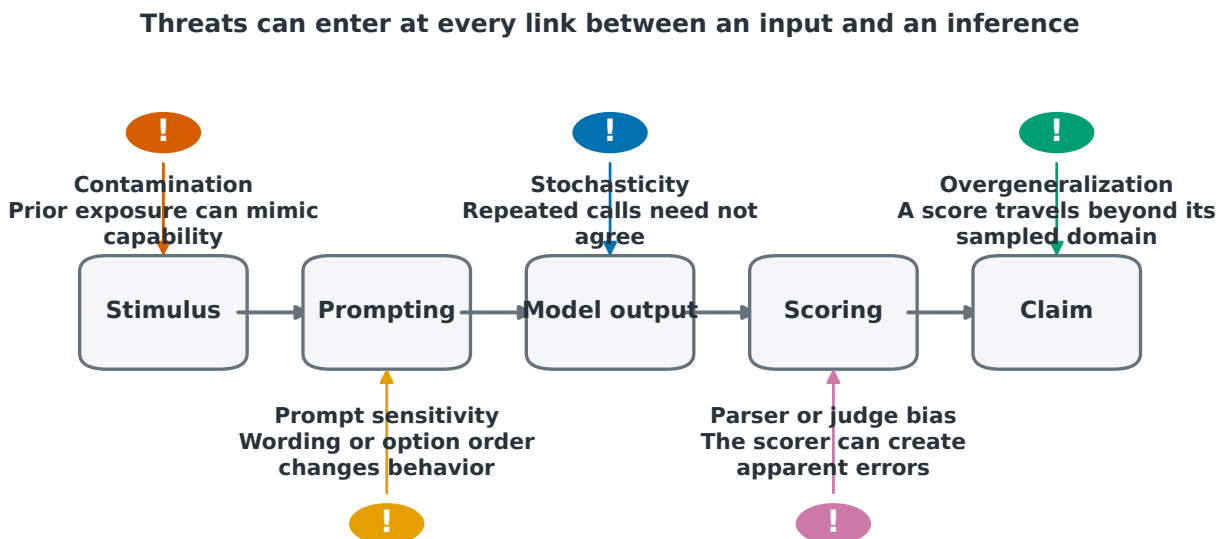
This descriptive consistency is encouraging evidence that no single plotting theme determined the overall result, but six synthetic styles are not a sample of all operational graphics. Curve-family profiles differed more sharply. GLM performed best on item-information and unfolding tasks, whereas Gemma’s advantage was concentrated in dichotomous and polytomous tasks.



**Figure 13:** Mean task-normalized scores by rendering profile and curve family. Similar style ranges coexist with larger content-profile differences, illustrating why both nuisance-condition and substantive-domain disaggregation are needed.

## 6 Validity Threats in LLM Benchmarking

Figure 14 locates five threats along the path from stimulus to claim. The diagram is deliberately causal in spirit: an observed score can be altered by factors introduced before, during, or after model generation.



**Figure 14:** Threats to benchmark inference. Contamination affects the relation between stimulus and capability; prompt sensitivity affects elicitation; stochasticity affects observed output; parser or judge bias affects scoring; overgeneralization affects the interpretation attached to the score.

### 6.1 Contamination and shortcut learning

Prior exposure can occur at the level of task instructions, raw text or images, labels, code, or public discussions (Sainz et al., 2023). Programmatic generation reduces exact-item exposure but does not prevent a model from learning the generating template. Designers should therefore vary parameters and rendering conditions, keep some generators or forms private where appropriate, record release dates, and avoid claims that require proving complete non-exposure.

Shortcuts can arise even without contamination. A model may infer the answer from file-

names, alt text, legend labels, repeated color mappings, or template regularities. Benchmark packaging should remove answer-bearing metadata from the model-visible input. Counterbalanced and adversarial variants can test whether a purported capability survives removal of the cue.

## **6.2 Prompt and interface sensitivity**

The prompt determines what behavior is elicited. A model can fail because it lacks the target capability, misunderstands an instruction, cannot emit the requested format, or follows a conflicting system policy. These are different findings. Reporting multiple plausible prompts and invalid-output rates helps distinguish them. A benchmark intended to compare model products should also treat provider preprocessing and tool access as part of the product, while a benchmark intended to compare base capabilities may disable them.

## **6.3 Stochasticity and version drift**

A single API response is one draw from a configured system. Repeated runs reveal whether model rankings are stable or depend on sampling. Hosted systems may change without a new public name. Access dates, model identifiers, raw outputs, and periodic anchor-item reruns are therefore essential. Historical scores should not be pooled across silent model versions as if they were measurements of one fixed object.

## **6.4 Scoring and judge error**

Parsers can mark semantically correct outputs wrong, while permissive extraction can reward extraneous or contradictory text. LLM judges may exhibit task-specific disagreement with people and each other (Bavaresco et al., 2025). Deterministic scoring should be preferred when ground truth is structured. When judgment is necessary, designers should publish the rubric, blind or randomize comparison order, estimate agreement with relevant human

experts, retain individual judgments, and specify adjudication.

## 6.5 Overgeneralization and consequential validity

Benchmark results often travel farther than their design. A score on synthetic IRT plots does not establish competence with empirical calibration reports, psychometric decision making, or the consequences of wrong advice. Raji and colleagues’ critique of “general” benchmarks is a reminder that dataset breadth is not a license for unbounded claims (Raji et al., 2021). The defensible interpretation should name the sampled content, model configuration, score rule, date, and intended use. Any model-selection decision should add use-specific validation rather than rely on a broad leaderboard alone.

# 7 A Reusable Recipe for Social Science Benchmark Projects

For a social science team beginning an LLM evaluation, the following sequence is a practical minimum:

1. Write one paragraph naming the decision, users, model system, context, and claims that are out of scope.
2. Create a construct map with narrow capabilities and observable behaviors; have subject-matter experts review it.
3. Build a content matrix crossing capabilities with domains, task types, difficulty, and nuisance transformations.
4. Choose an item source and ground-truth process; retain provenance and separate target-model generation from answer creation.
5. Pilot prompts and response schemas on several model families; freeze administration rules before comparison.
6. Define form assembly and analysis denominators, including invalid outputs and refusals.

7. Implement scoring with unit tests, diagnostic baselines, and human validation for judgment-based components.
8. Report disaggregated results and uncertainty; inspect errors before ranking models.
9. Publish documentation, raw outputs where permitted, code, version identifiers, known limitations, and maintenance plans.
10. Refresh, revalidate, or retire when exposure, saturation, model drift, or new intended uses undermine the original argument.

This recipe is intentionally closer to assessment design than to downloading a test set and calculating accuracy. It also remains different from human psychometrics: a team need not fit an IRT model to every benchmark or treat models as people. The transferable insight is evidentiary discipline. Define the claim, design observations that bear on it, standardize the elicitation, quantify uncertainty, and constrain the interpretation.

## 8 Discussion

LLM benchmarking and social science measurement have much to offer one another. Benchmark developers contribute practices for executable evaluation, large-scale automation, adversarial testing, and rapid iteration. Social scientists contribute theories of constructs, response processes, validity arguments, fairness, sampling, and consequences. ECBD demonstrates that this exchange is already possible (Liu et al., 2024). BetterBench’s lifecycle analysis similarly shows that benchmark quality depends on design, implementation, documentation, maintenance, and eventual retirement rather than predictive performance alone (Reuel et al., 2024).

IRT-Viz-Eval illustrates the benefits of this combined perspective. Programmatic generation provides exact answers; factorial rendering supports controlled invariance evidence; structured records preserve provenance; field-level scoring supports error analysis; and diagnostic baselines test mechanics. The two real-model administrations add a second lesson.

An overall near-tie concealed opposite strengths by task type, a 16.7% empty-output rate for one routed system, a sixfold difference in median latency, and a different ordering under raw-point rather than task-normalized aggregation. Benchmark construction decisions were therefore not background details; they changed what a comparison would mean.

The validity argument remains bounded. The benchmark samples clean synthetic plots, only two routed systems and one response per task were observed, provider defaults were not fully harmonized, the rationale rule needs expert validation, and the six style profiles do not represent the ecology of operational psychometric graphics. The cluster-bootstrap intervals reflect variation across 14 mathematical base stimuli, not model stochasticity, future provider drift, or uncertainty in scoring. In particular, the empirical results do not establish that either model is suitable for unsupervised psychometric decisions.

The next validation phase should repeat administrations, counterbalance plausible prompts, and obtain blinded expert scores for a stratified response sample. Content extensions should include empirical estimation noise, confidence bands, differential item functioning plots, item-fit diagnostics, score-report visualizations, and accessibility conditions. An external study could then test whether benchmark performance predicts quality on authentic psychometric review tasks.

## 9 Conclusion

An LLM benchmark is best understood as a maintained measurement system. Its data are only one component. The prompts define elicitation, the model configuration defines the object, the parser or judge defines the observation, the assembly rule defines the sampled domain, and the validity argument defines what the score can mean. These features distinguish benchmarking from conventional educational and psychological assessment while preserving a shared commitment to evidence-based interpretation.

For social scientists, the practical lesson is straightforward: do not begin or end with a

leaderboard. Begin with an intended use and a defensible capability claim. Build backward to the evidence and tasks, then move forward through standardized administration, scoring, validation, reporting, and maintenance. The empirical demonstration showed why: two systems with the same rounded overall score behaved differently across capabilities, output validity, latency, and aggregation rules. IRT-Viz-Eval provides one complete implementation of this process and a bounded foundation for studying multimodal interpretation of psychometric graphics.

# A Benchmark Construction Checklist

**Table 6:** Questions to answer before releasing or using an LLM benchmark.

| Stage        | Required questions                                                                                                                            |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Intended use | Who will use the result? What decision will it inform? What interpretations and uses are explicitly unsupported?                              |
| Object       | What exact model, version, interface, tools, retrieval sources, and provider settings are being evaluated? Can these change during the study? |
| Capability   | What narrow behavior is claimed? Which alternative processes could produce the same observed answer?                                          |
| Content      | What domains, task types, difficulty levels, languages, groups, and modalities are sampled? Which cells are absent?                           |
| Ground truth | Who or what creates the reference answer? Is it independently checked? How are ambiguity and disagreement represented?                        |
| Adaptation   | What system prompt, user prompt, examples, image transformations, output schema, decoding settings, and retries are used?                     |
| Assembly     | How are tasks selected and weighted? Are forms identical across models? What seed and exclusion rules are used?                               |
| Scoring      | How are raw outputs parsed? How are refusals, multiple answers, malformed outputs, and missing fields scored?                                 |
| Judgment     | If people or LLMs judge outputs, what rubric, training, blinding, agreement, adjudication, and judge provenance are used?                     |
| Uncertainty  | What varies across items, repeated calls, prompts, forms, models, and judges? Which interval or model represents that variation?              |
| Robustness   | Which meaning-preserving changes should leave scores stable? Which difficulty manipulations should change scores predictably?                 |

| Stage         | Required questions                                                                                                        |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| Contamination | Could items, answers, templates, or code have entered training data?<br>What evidence and mitigation are available?       |
| Documentation | Are data provenance, licenses, model versions, prompts, raw outputs, code, exclusions, costs, and limitations documented? |
| Maintenance   | Who owns updates? How will exposure, drift, and saturation be monitored? What triggers a new version or retirement?       |

## B IRT-Viz-Eval Task Families

**Table 7:** Current task families and primary evidence fields.

| Task family          | Observable response                               | Primary scoring                          |
|----------------------|---------------------------------------------------|------------------------------------------|
| Model identification | IRT model or curve family plus visual rationale   | Exact categorical match + rationale      |
| Parameter estimation | $a, b, c, d$ where applicable                     | Numeric tolerance bands                  |
| Probability reading  | $P(\theta)$ at a specified location               | Numeric tolerance bands                  |
| Information peak     | Peak location and height                          | Numeric tolerance bands                  |
| Category reasoning   | Category count, modal category, expected score    | Integer, categorical, and numeric fields |
| Test characteristic  | Form length and expected raw score                | Integer and numeric fields               |
| Non-monotonic peak   | Non-monotonicity, peak location, peak probability | Categorical and numeric fields           |

## C Linked Data Schema

Each line is a JSON object with a stable identifier. The joins are:

`stimulus_id → task_id → response_id → judgment_id.`

**Table 8:** Minimum fields retained in each benchmark artifact.

| Artifact | Core fields                                                                                                                                                            |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stimulus | Base and rendered identifiers; model family; mathematical parameters; ground truth; coordinate path; image path; style, axes, palette, legend, size, and render engine |
| Task     | Task type; prompt; prompt-template identifier; expected answer fields; field types; numeric thresholds; rubric; image link                                             |
| Response | Target model and version; system and user prompts; decoding settings; seed; access time; latency; raw output; parsed output; parser version                            |
| Judgment | Response link; field-level errors and scores; rationale score; total and maximum score; normalized score; judge type, version, prompt, and timestamp                   |

## D Reporting Template for an Empirical Model Study

An empirical paper using this benchmark should report:

1. benchmark release, generator version, date generated, item count, task count, and form seed;
2. every target model’s provider, identifier, access date, modality support, context limit, and tool configuration;
3. full system and user prompts, examples, response schema, image preprocessing, decoding, retries, and repetitions;
4. primary and secondary outcomes, aggregation weights, invalid-output policy, uncertainty procedure, and multiplicity handling;
5. deterministic scorer version and, if applicable, judge model, prompt, order randomization, human validation sample, and agreement;
6. overall and disaggregated results by task, curve family, style, and base stimulus, including denominators and confidence intervals;
7. parse failures, refusals, latency, token use, cost, exclusions, and missing data;
8. contamination considerations, benchmark limitations, intended interpretation, unsupported uses, and a link to raw outputs where licensing permits.

## E Reproducibility Resources

The repository includes a one-command local pipeline, source modules for psychometric generation and rendering, JSON schemas, target and judge prompts, generated stimuli and task records, diagnostic responses and judgments, analysis tables, and the script that generated every figure in this article. The local command is:

```
bash scripts/run_local_benchmark.sh
```

After installing the optional provider dependency and authenticating locally, a small Hugging Face pilot can be administered with:

```
PYTHONPATH=src python3 -m irt_viz_eval.cli run-huggingface \  
  --models MODEL_ID:PROVIDER --limit 10 \  
  --request-timeout 90 --max-retries 1
```

The adapter supports task limits, repeated administrations, resumable output, bounded timeouts, transient retries, and per-task progress. The repository also includes an OpenAI Responses API adapter; the shared response schema permits future Gemini or local-runtime adapters without changing the task and judgment records. Model identifiers are supplied at runtime because provider availability changes. Credentials are never stored in the repository. The empirical preparation script deduplicates response identifiers, requires complete 198-task coverage, applies the deterministic scorer, and produces the published tables and cluster-bootstrap intervals.

The paper figures can be regenerated with:

```
python3 paper/generate_figures.py
```

The complete empirical outputs used in this manuscript can be regenerated after the two raw response files are present with:

```
PYTHONPATH=src python3 paper/prepare_empirical_results.py
```

Downloaded source articles used during manuscript development are not redistributed. Persistent publisher and repository locations are supplied in the reference list where available.

## F Declarations

**Data and code availability.** An anonymized archive containing the benchmark generator, task records, schemas, prompts, scoring code, deduplicated model responses and judgments, diagnostic outputs, tests, analysis tables, and figure-generation scripts accompanies the review submission. The public repository URL will be inserted after double-blind review.

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